Exercise Series 8 (Solutions)

Exercise 1 (a) $L(\theta) = L(\theta|\underline{x}) = \prod_{i=1}^{n} f_\theta(x_i)$

$$= \prod_{i=1}^{n} \frac{e^{-(x_i-\theta)}}{\{1+e^{-(x_i-\theta)}\}^2}, \ \theta \in \mathbb{R}$$

log-likelihood :

$$\ell(\theta) = \ell(\theta|\underline{x}) = \log L(\theta|\underline{x}) = \sum_{i=1}^{n} \log\left[\frac{e^{-(x_i-\theta)}}{\{1+e^{-(x_i-\theta)}\}^2}\right]$$

$$= \sum_{i=1}^{n}\left[-(x_i-\theta) - 2\log\{1+e^{-(x_i-\theta)}\}\right], \ \theta \in \mathbb{R}.$$

$$\ell'(\theta) = \sum_{i=1}^{n} \frac{\partial}{\partial\theta}\left[-(x_i-\theta) - 2\log\{1+e^{-(x_i-\theta)}\}\right]$$

$$\frac{\partial}{\partial\theta}\left[-(x_i-\theta) - 2\log\{1+e^{-(x_i-\theta)}\}\right] = 1 - 2\frac{e^{-(x_i-\theta)}}{1+e^{-(x_i-\theta)}}$$

$$\therefore \ \ell'(\theta) = \sum_{i=1}^{n}\left\{1 - 2\frac{e^{-(x_i-\theta)}}{1+e^{-(x_i-\theta)}}\right\} = n - 2\sum_{i=1}^{n}\frac{e^{-(x_i-\theta)}}{1+e^{-(x_i-\theta)}}$$

likelihood equation:

$$\ell'(\theta) = 0 \iff \sum_{i=1}^{n}\frac{e^{-(x_i-\theta)}}{1+e^{-(x_i-\theta)}} = \frac{n}{2}$$

(b) $$\ell''(\theta) = \sum_{i=1}^{n}\frac{\partial^2}{\partial\theta^2}\left[-(x_i-\theta) - 2\log\{1+e^{-(x_i-\theta)}\}\right]$$

$$= \sum_{i=1}^{n}\frac{\partial}{\partial\theta}\left[1 - 2\frac{e^{-(x_i-\theta)}}{1+e^{-(x_i-\theta)}}\right]$$

$$\frac{\partial}{\partial\theta}\left[1 - 2\frac{e^{-(x_i-\theta)}}{1+e^{-(x_i-\theta)}}\right] = -2\frac{e^{-(x_i-\theta)}}{\{1+e^{-(x_i-\theta)}\}^2}$$

$$\therefore \; l''(\theta) = -2 \sum_{i=1}^{n} \frac{e^{-(x_i-\theta)}}{\{1+e^{-(x_i-\theta)}\}^2}$$

$$I_n(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2} l(\theta|\underline{X})\right] = 2\sum_{i=1}^{n} \mathbb{E}_\theta\left[\frac{e^{-(X_i-\theta)}}{\{1+e^{-(X_i-\theta)}\}^2}\right]$$

$$= 2n \; \mathbb{E}_\theta\left[\frac{e^{-(X_1-\theta)}}{\{1+e^{-(X_1-\theta)}\}^2}\right]$$

Now, $$\mathbb{E}_\theta\left[\frac{e^{-(X_1-\theta)}}{\{1+e^{-(X_1-\theta)}\}^2}\right] = \int_{-\infty}^{\infty} \frac{e^{-(x-\theta)}}{\{1+e^{-(x-\theta)}\}^2} \frac{e^{-(x-\theta)}}{\{1+e^{-(x-\theta)}\}^2} dx$$

$$= \int_{-\infty}^{\infty} \frac{e^{-2(x-\theta)}}{\{1+e^{-(x-\theta)}\}^4} dx$$

$$= \int_{-\infty}^{\infty} \frac{e^{-2z}}{(1+e^{-z})^4} dz \qquad (x-\theta = z)$$

$$= \int_{1}^{\infty} \frac{t-1}{t^4} dt \qquad \begin{aligned} 1+e^{-z} &= t \\ -e^{-z}dz &= dt \end{aligned} \qquad \begin{array}{c|c|c} z & -\infty & \infty \\ \hline t & \infty & 1 \end{array}$$

$$= \int_{1}^{\infty} \frac{dt}{t^3} - \int_{1}^{\infty} \frac{dt}{t^4} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

$$\therefore \; I_n(\theta) = 2n \frac{1}{6} = \frac{n}{3}$$

$$I(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2} l(\theta|\underline{x})\right] = 2 \cdot \frac{1}{6} = \frac{1}{3}$$

(c) Fisher's scoring equation:

$$\theta_{n+1} = \theta_n + \frac{l'(\theta_n)}{I_n(\theta_n)} = \theta_n + \frac{n - 2\sum_{i=1}^{n} \frac{e^{-(x_i-\theta_n)}}{1+e^{-(x_i-\theta_n)}}}{n/3}$$

$$= \theta_n + 3 - \frac{6}{n} \sum_{i=1}^{n} \frac{e^{-(x_i-\theta_n)}}{1+e^{-(x_i-\theta_n)}}$$

See attached codes for parts (d) – (g).

To use the iterative algorithm, one important issue is the choice of the initial value θ_0.

Here, we observe that θ is the mean as well as the median of X [The dist. of X is symmetric about θ]

So, we may use the sample mean or the sample median as our starting value.

Exercise 2 (a) $\dfrac{e^{\alpha+\beta x}}{1+e^{\alpha+\beta x}} = 1 - \dfrac{1}{1+e^{\alpha+\beta x}}$.

For fixed α, $\beta \longmapsto \alpha+\beta x$ is $\uparrow$ in β

$\therefore\ \beta \longmapsto e^{\alpha+\beta x}$ is $\uparrow$ in β

$\therefore\ \beta \longmapsto 1+e^{\alpha+\beta x}$ is $\uparrow$ in β

$\therefore\ \beta \longmapsto \dfrac{1}{1+e^{\alpha+\beta x}}$ is $\downarrow$ in β

$\therefore\ \beta \longmapsto 1 - \dfrac{1}{1+e^{\alpha+\beta x}}$ is $\uparrow$ in β

The other case (fixed β) is similar.

(b) Our model is: $Y_i \overset{\text{indep.}}{\sim} \text{Bernoulli}(p_i)$, $i=1,\dots,n$,

where $p_i = \dfrac{e^{\alpha+\beta x_i}}{1+e^{\alpha+\beta x_i}} =: p_i(\alpha,\beta)$, say.

The likelihood is:

$$L(\alpha,\beta) = \prod_{i=1}^{n} f_{\alpha,\beta}(y_i)$$

$$\begin{aligned} f_{\alpha,\beta}(y_i) &= p_i(\alpha,\beta)^{y_i}\{1-p_i(\alpha,\beta)\}^{1-y_i} \\ &= \left(\frac{e^{\alpha+\beta x_i}}{1+e^{\alpha+\beta x_i}}\right)^{y_i}\left(\frac{1}{1+e^{\alpha+\beta x_i}}\right)^{1-y_i} \\ &= \frac{e^{y_i(\alpha+\beta x_i)}}{1+e^{\alpha+\beta x_i}} \end{aligned}$$

$$\therefore\ L(\alpha,\beta) = \prod_{i=1}^{n} \frac{e^{y_i(\alpha+\beta x_i)}}{1+e^{\alpha+\beta x_i}}, \quad \alpha,\beta \in \mathbb{R}$$

likelihood equations:

$$\left.\begin{aligned}\frac{\partial}{\partial\alpha}\ell(\alpha,\beta) &= 0\\ \frac{\partial}{\partial\beta}\ell(\alpha,\beta) &= 0\end{aligned}\right\} \qquad \begin{aligned}\ell(\alpha,\beta) &= \log L(\alpha,\beta)\\ &= \sum_{i=1}^{n}\left\{y_i(\alpha+\beta x_i) - \log\left(1+e^{\alpha+\beta x_i}\right)\right\}\end{aligned}$$

$$\frac{\partial}{\partial\alpha}\ell(\alpha,\beta) = \sum_{i=1}^{n}\left\{y_i - \frac{e^{\alpha+\beta x_i}}{1+e^{\alpha+\beta x_i}}\right\}$$

$$\frac{\partial}{\partial\beta}\ell(\alpha,\beta) = \sum_{i=1}^{n}\left\{x_i y_i - \frac{x_i e^{\alpha+\beta x_i}}{1+e^{\alpha+\beta x_i}}\right\}$$

$\therefore$ likelihood equations are:

$$\left.\begin{aligned}\sum_{i=1}^{n}\frac{e^{\alpha+\beta x_i}}{1+e^{\alpha+\beta x_i}} &= \sum_{i=1}^{n} y_i\\ \sum_{i=1}^{n} x_i\frac{e^{\alpha+\beta x_i}}{1+e^{\alpha+\beta x_i}} &= \sum_{i=1}^{n} x_i y_i\end{aligned}\right\}$$

(c) In order to find the Fisher's scoring equations, we need to find the information matrix $I_n(\alpha,\beta)$.

This will be a 2×2 matrix.

In this model, Y_i's are independent, but not identical.

Therefore, $I_n(\theta) = \sum_{i=1}^{n} I_i(\theta)$, where

$$I_i(\alpha,\beta) = -\begin{bmatrix}\mathbb{E}\left[\frac{\partial^2}{\partial\alpha^2}\ell(\alpha,\beta\,|\,Y_i)\right] & \mathbb{E}\left[\frac{\partial^2}{\partial\beta\partial\alpha}\ell(\alpha,\beta\,|\,Y_i)\right]\\ \text{symmetry} & \mathbb{E}\left[\frac{\partial^2}{\partial\beta^2}\ell(\alpha,\beta\,|\,Y_i)\right]\end{bmatrix}$$

$$\frac{\partial}{\partial \alpha} l(\alpha, \beta \mid y_i) = y_i - \frac{e^{\alpha+\beta x_i}}{1+e^{\alpha+\beta x_i}}$$

$$\frac{\partial}{\partial \beta} l(\alpha, \beta \mid y_i) = x_i y_i - x_i \frac{e^{\alpha+\beta x_i}}{1+e^{\alpha+\beta x_i}}$$

$$\frac{\partial^2}{\partial \alpha^2} l(\alpha, \beta \mid y_i) = - \frac{e^{\alpha+\beta x_i}}{\{1+e^{\alpha+\beta x_i}\}^2}$$

$$\frac{\partial^2}{\partial \beta \partial \alpha} l(\alpha, \beta \mid y_i) = - x_i \frac{e^{\alpha+\beta x_i}}{\{1+e^{\alpha+\beta x_i}\}^2}$$

$$\frac{\partial^2}{\partial \beta^2} l(\alpha, \beta \mid y_i) = - x_i^2 \frac{e^{\alpha+\beta x_i}}{\{1+e^{\alpha+\beta x_i}\}^2}$$

terms free of y_i

$$\therefore \; I_i(\theta) = I_i(\alpha, \beta) = \frac{e^{\alpha+\beta x_i}}{\{1+e^{\alpha+\beta x_i}\}^2} \begin{bmatrix} 1 & x_i \\ x_i & x_i^2 \end{bmatrix}$$

$$I_n(\theta) = \sum_{i=1}^{n} I_i(\theta) = \sum_{i=1}^{n} \frac{e^{\alpha+\beta x_i}}{\{1+e^{\alpha+\beta x_i}\}^2} \begin{bmatrix} 1 & x_i \\ x_i & x_i^2 \end{bmatrix}$$

Fisher's scoring equation:

$$\begin{pmatrix} \alpha_{n+1} \\ \beta_{n+1} \end{pmatrix} = \begin{pmatrix} \alpha_n \\ \beta_n \end{pmatrix} + I_n(\alpha_n, \beta_n)^{-1} \begin{pmatrix} \frac{\partial}{\partial \alpha} l(\alpha, \beta) \\ \frac{\partial}{\partial \beta} l(\alpha, \beta) \end{pmatrix} \Bigg|_{\substack{\alpha = \alpha_n \\ \beta = \beta_n}}$$

$$= \begin{pmatrix} \alpha_n \\ \beta_n \end{pmatrix} + I_n(\alpha_n, \beta_n)^{-1} \begin{pmatrix} \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \frac{e^{\alpha_n+\beta_n x_i}}{1+e^{\alpha_n+\beta_n x_i}} \\ \sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \frac{e^{\alpha_n+\beta_n x_i}}{1+e^{\alpha_n+\beta_n x_i}} \end{pmatrix}$$

Since I_n is a 2×2 matrix, it is possible to find its inverse in a closed-form.

Exercise 3 (a) $T = \sum_{i=1}^{n} a_i X_i$, $X_1, \ldots, X_n$ iid (μ, σ^2)

$$\mathbb{E}_\theta(T) = \sum_{i=1}^{n} a_i \mathbb{E}_\theta(X_i) = \sum_{i=1}^{n} a_i \mu \quad [\theta = (\mu, \sigma^2)]$$

$$= \mu \sum_{i=1}^{n} a_i$$

T is unbiased for μ $\iff \mu \sum_{i=1}^{n} a_i = \mu \quad \forall \mu$

$$\iff \sum_{i=1}^{n} a_i = 1$$

(b) $$\mathrm{var}_\theta(T) = \mathrm{var}_\theta\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} \mathrm{var}_\theta(a_i X_i) \quad [\because X_i\text{'s are indep}]$$

$$= \sum_{i=1}^{n} a_i^2 \,\mathrm{var}_\theta(X_i)$$

$$= \sum_{i=1}^{n} a_i^2 \sigma^2 \quad [\because \mathrm{var}_\theta(X_i) = \sigma^2 \ \forall i]$$

$$= \sigma^2 \sum_{i=1}^{n} a_i^2$$

(c) To find the unb. estimator $\sum_{i=1}^{n} a_i X_i$ with min. variance, we need to minimize $\sigma^2 \sum_{i=1}^{n} a_i^2$ s.t. $\sum_{i=1}^{n} a_i = 1$

$\iff$ minimize $\sum_{i=1}^{n} a_i^2$ s.t. $\sum_{i=1}^{n} a_i$

Using Cauchy-Schwarz inequality,

$$\left(\sum_{i=1}^{n} a_i\right)^2 \leq n \sum_{i=1}^{n} a_i^2 \implies \sum_{i=1}^{n} a_i^2 \geq \frac{1}{n}\left(\sum_{i=1}^{n} a_i\right)^2$$

$$= \frac{1}{n}$$

The min is achieved, i.e., "=" holds iff $(a_1, \ldots, a_n) \propto (1, \ldots, 1)$, i.e, $a_i = c \ \forall i$

Since $\sum_{i=1}^{n} a_i = 1$, we set $c = \frac{1}{n}$.

$\therefore$ The estimator of the form (a) with min. variance is $\frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$.

Exercise 4 (a) $f_p(x) = p^x(1-p)^{1-x}$, $x \in \{0,1\}$

$$\log f_p(x) = x\log p + (1-x)\log(1-p)$$

$$\frac{\partial}{\partial p}\log f_p(x) = \frac{x}{p} - \frac{1-x}{1-p}$$

$$\frac{\partial^2}{\partial p^2}\log f_p(x) = -\frac{x}{p^2} - \frac{1-x}{(1-p)^2}$$

$$\therefore I(p) = -\mathbb{E}_p\left[\frac{\partial^2}{\partial p^2}\log f_p(X)\right] = \mathbb{E}_p\left[\frac{X}{p^2} + \frac{1-X}{(1-p)^2}\right]$$

$$= \frac{p}{p^2} + \frac{1-p}{(1-p)^2}$$

$$= \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}$$

$$\therefore I_n(p) = \frac{n}{p(1-p)}.$$

CRLB for UE of p : $\dfrac{p(1-p)}{n}$.

MLE $= \bar{X}$ attains the CRLB.

(b) $f_\lambda(x) = e^{-\lambda}\dfrac{\lambda^x}{x!}$, $x = 0, 1, \ldots$

$$\log f_\lambda(x) = -\lambda + x\log\lambda - \log(x!)$$

$$\frac{\partial}{\partial\lambda}\log f_\lambda(x) = -1 + \frac{x}{\lambda}$$

$$\frac{\partial^2}{\partial\lambda^2}\log f_\lambda(x) = -\frac{x}{\lambda^2}$$

$$I(\lambda) = -\mathbb{E}_\lambda\left[-\frac{X}{\lambda^2}\right] = \mathbb{E}_\lambda\left(\frac{X}{\lambda^2}\right) = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$$

$I_n(\lambda) = \dfrac{n}{\lambda}$. CRLB for UE of $\lambda = \dfrac{\lambda}{n}$

MLE $\bar{X}$ attains the CRLB.

(c) $f_p(x) = p(1-p)^x \quad , \quad x = 0, 1, \ldots$

$\log f_p(x) = \log p + x \log(1-p)$

$\frac{\partial}{\partial p} \log f_p(x) = \frac{1}{p} - \frac{x}{1-p}$

$\frac{\partial^2}{\partial p^2} \log f_p(x) = -\frac{1}{p^2} - \frac{x}{(1-p)^2}$

$I(p) = \frac{1}{p^2} + \frac{1}{(1-p)^2} \mathbb{E}_p(X) = \frac{1}{p^2} + \frac{1}{(1-p)^2} \frac{1-p}{p}$

$= \frac{1}{p^2} + \frac{1}{p(1-p)} = \frac{1}{p^2(1-p)}$

CRLB : $\frac{p^2(1-p)}{n}$. MLE $= \frac{1}{\bar{X}+1}$

Calculating the variance of the MLE is difficult.

But, $\sqrt{n}\left(\bar{X} - \frac{1-p}{p}\right) \xrightarrow{\mathcal{D}} N\left(0, \frac{1-p}{p^2}\right)$. Using delta method, it can be shown that $\sqrt{n}\left(\frac{1}{\bar{X}+1} - p\right) \xrightarrow{\mathcal{D}} N(0, p^2(1-p))$

So, the asymptotic variance of the MLE matches the CRLB.

(d) $f_\theta(x) = \frac{1}{\theta} \quad , \quad 0 \leq x \leq \theta$

$\log f_\theta(x) = -\log \theta$

$\frac{\partial}{\partial \theta} \log f_\theta(x) = -\frac{1}{\theta} \quad , \quad \frac{\partial^2}{\partial \theta^2} \log f_\theta(x) = \frac{1}{\theta^2}$

$I(\theta) = \mathbb{E}_\theta\left[\left\{\frac{\partial}{\partial \theta} \log f_\theta(X)\right\}^2\right] = \frac{1}{\theta^2}$

However, $-\mathbb{E}_\theta\left[\frac{\partial^2}{\partial \theta^2} \log f_\theta(X)\right] = -\frac{1}{\theta^2} < 0$

Check that, in this example,

$\frac{d}{d\theta} \mathbb{E}_\theta\left[\log f_\theta(X)\right] \neq \mathbb{E}_\theta\left[\frac{\partial}{\partial \theta} \log f_\theta(X)\right]$

Here, MLE $= X_{(n)}$. It was shown in the class that $\text{var}\left(X_{(n)}\right)$ is of the order $\frac{1}{n^2}$, which is much smaller than the CRLB.

Here, the CRLB is invalid since the regularity condition do not hold.

(e) $f_\mu(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2}}$

$$\log f_\mu(x) = -\frac{1}{2}\log(2\pi) - \frac{(x-\mu)^2}{2}$$

$$\frac{\partial}{\partial\mu}\log f_\mu(x) = \frac{1}{2}\cdot 2(x-\mu) = x-\mu$$

$$\frac{\partial^2}{\partial\mu^2}\log f_\mu(x) = -1$$

$\therefore I(\mu) = 1$. CRLB $= \frac{1}{n}$

MLE $\bar{X}$ attains the CRLB.

(f) $f_\sigma(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{x^2}{2\sigma^2}}$

Define $\sigma^2 = \theta$. Then,

$$f_\theta(x) = \frac{1}{\sqrt{2\pi\theta}} e^{-\frac{x^2}{2\theta}}$$

$$\log f_\theta(x) = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\theta - \frac{x^2}{2\theta}$$

$$\frac{\partial}{\partial\theta}\log f_\theta(x) = -\frac{1}{2\theta} + \frac{x^2}{2\theta^2}$$

$$\frac{\partial^2}{\partial\theta^2}\log f_\theta(x) = \frac{1}{2\theta^2} - \frac{x^2}{\theta^3}$$

$$I(\sigma^2) = I(\theta) = \frac{1}{\theta^3}\mathbb{E}_\theta(X^2) - \frac{1}{2\theta^2}$$

$$= \frac{1}{\theta^3}\cdot\theta - \frac{1}{2\theta^2} = \frac{1}{2\theta^2} = \frac{1}{2\sigma^4}.$$

$$\text{CRLB} = \frac{2\sigma^4}{n}$$

$\text{MLE} = \frac{1}{n}\sum_{i=1}^{n} X_i^2$ attains the CRLB.

(g) $$f_{\mu,\sigma}(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Reparameterize $\theta = \sigma^2$ and write

$$f_{\mu,\theta}(x) = \frac{1}{\sqrt{2\pi\theta}} e^{-\frac{(x-\mu)^2}{2\theta}}$$

$$\log f_{\mu,\theta}(x) = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\theta - \frac{(x-\mu)^2}{2\theta}$$

$$\frac{\partial}{\partial\mu}\log f_{\mu,\theta}(x) = \frac{x-\mu}{\theta}, \quad \frac{\partial}{\partial\theta}\log f_{\mu,\theta}(x) = -\frac{1}{2\theta} + \frac{(x-\mu)^2}{2\theta^2}$$

$$\frac{\partial^2}{\partial\mu^2}\log f_{\mu,\theta}(x) = -\frac{1}{\theta}, \quad \frac{\partial^2}{\partial\theta^2}\log f_{\mu,\theta}(x) = \frac{1}{2\theta^2} - \frac{(x-\mu)^2}{\theta^3}$$

$$\frac{\partial^2}{\partial\theta\partial\mu}\log f_{\mu,\theta}(x) = \frac{\partial^2}{\partial\mu\partial\theta}\log f_{\mu,\theta}(x) = -\frac{x-\mu}{\theta^2}.$$

$$\therefore I(\mu,\theta) = \begin{bmatrix} \frac{1}{\theta} & 0 \\ 0 & \frac{1}{2\theta^2} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{bmatrix}$$

$$I(\mu,\sigma^2) = \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{bmatrix}$$

MLE of μ is $\bar{X}$

σ^2 is $S^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$

$$\operatorname{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \frac{nS^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$\Rightarrow \mathbb{E}\left(\frac{nS^2}{\sigma^2}\right) = n-1$$

$$\operatorname{var}\left(\frac{nS^2}{\sigma^2}\right) = 2(n-1)$$

$$\therefore \operatorname{var}(S^2) = \frac{2(n-1)}{n^2}\sigma^4$$

. $\bar{X}$ & S^2 are independent.

CRLB for UE of (μ, σ^2) is

$$V_1 = \begin{bmatrix} \frac{\sigma^2}{n} & 0 \\ 0 & \frac{2\sigma^4}{n} \end{bmatrix}$$

var of $\begin{pmatrix} \bar{X} \\ S^2 \end{pmatrix}$ is $V_2 = \begin{bmatrix} \frac{\sigma^2}{n} & 0 \\ 0 & \frac{2\sigma^4}{n} - \frac{2\sigma^4}{n^2} \end{bmatrix}$

$$V_1 - V_2 = \begin{bmatrix} 0 & 0 \\ 0 & \frac{2\sigma^4}{n} \end{bmatrix} \Rightarrow V_1 \succ V_2$$

Notice that S^2 is not an UE of σ^2.

(h) $f_\sigma(x) = \frac{1}{2\sigma} e^{-\frac{|x|}{\sigma}}$

$$\log f_\sigma(x) = -\log 2 - \log \sigma - \frac{|x|}{\sigma}$$

$$\frac{\partial}{\partial\sigma} \log f_\sigma(x) = -\frac{1}{\sigma} + \frac{|x|}{\sigma^2}$$

$$\frac{\partial^2}{\partial\sigma^2} \log f_\sigma(x) = \frac{1}{\sigma^2} - \frac{2|x|}{\sigma^3}$$

$$\therefore I(\sigma) = \frac{2}{\sigma^3} \mathbb{E}_\sigma(|X|) - \frac{1}{\sigma^2}$$

$$= \frac{2}{\sigma^3}\sigma - \frac{1}{\sigma^2} = \frac{1}{\sigma^2}.$$

CRLB $= \frac{\sigma^2}{n}$. MLE $= \frac{1}{n}\sum_{i=1}^{n}|X_i|$ has variance $\frac{\sigma^2}{n}$.

So, it attains the CRLB.

(i) $f_\theta(x) = \frac{1}{\sqrt{2\pi}\,\theta} e^{-\frac{(x-\theta)^2}{2\theta^2}}$

$$\log f_\theta(x) = -\frac{1}{2}\log(2\pi) - \log\theta - \frac{(x-\theta)^2}{2\theta^2}$$

$$= -\frac{1}{2}\log(2\pi) - \log\theta - \frac{x^2}{2\theta^2} - \frac{1}{2} + \frac{x}{\theta}$$

$$\frac{\partial}{\partial\theta} \log f_\theta(x) = -\frac{1}{\theta} + \frac{x^2}{\theta^3} - \frac{x}{\theta^2}$$

$$\frac{\partial^2}{\partial\theta^2} \log f_\theta(x) = \frac{1}{\theta^2} - \frac{3x^2}{\theta^4} + \frac{2x}{\theta^3}$$

$$\therefore I(\theta) = \frac{3(\theta^2+\theta^2)}{\theta^4} - \frac{1}{\theta^2} - \frac{2\theta}{\theta^3} = \frac{3}{\theta^2}$$

CRLB $= \frac{\theta^2}{3n}$.

(j) $f_\theta(x) = \frac{1}{\sqrt{2\pi\theta}} e^{-\frac{(x-\theta)^2}{2\theta}}$

$$\log f_\theta(x) = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\theta - \frac{(x-\theta)^2}{2\theta}$$

$$= -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\theta - \frac{x^2}{2\theta} - \frac{\theta}{2} + x$$

$$\frac{\partial}{\partial\theta}\log f_\theta(x) = -\frac{1}{2\theta} + \frac{x^2}{2\theta^2} - \frac{1}{2}$$

$$\frac{\partial^2}{\partial\theta^2}\log f_\theta(x) = \frac{1}{2\theta^2} - \frac{x^2}{\theta^3}$$

$$I(\theta) = \frac{\mathbb{E}_\theta(X^2)}{\theta^3} - \frac{1}{2\theta^2}$$

$$= \frac{\theta+\theta^2}{\theta^3} - \frac{1}{2\theta^2} = \frac{1}{\theta^2} + \frac{1}{\theta} - \frac{1}{2\theta^2} = \frac{1}{\theta} + \frac{1}{2\theta^2}$$

(k) $f_\theta(x) = \theta e^{-\theta x}$

$$\log f_\theta(x) = \log\theta - \theta x$$

$$\frac{\partial}{\partial\theta}\log f_\theta(x) = \frac{1}{\theta} - x$$

$$\frac{\partial^2}{\partial\theta^2}\log f_\theta(x) = -\frac{1}{\theta^2}$$

$I(\theta) = \frac{1}{\theta^2}$. $\quad CRLB = \frac{\theta^2}{n}$.